LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	12
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Switches: - toggle switches, push switches, rocker button switches, rotary switches, micro switches, time s/w, mercury s/w, pressure s/w, thermal s/w, proximity s/w, rheostat and relays
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Switches: - toggle switches, push switches, rocker button switches, rotary switches, micro switches, time s/w, mercury s/w, pressure s/w, thermal s/w, proximity s/w, rheostat and relays. Safeties to be taken while installing switches & relays. Practical switches & relays.
5. Specific Objectives	13.1.2.1 Knowledge of Static converting equipment, Bus bars, cable terminations, Switches.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification and installation of various switch types; Safety precautions during installation; Instructor supervision of practical assembly
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a switch in an electrical circuit?
(b) Name three types of switches used in avionics systems.
(c) Why is safety critical when installing switches and relays?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Toggle switch operation and application • Push and rocker button switch mechanisms • Rotary switch configurations and uses • Micro switch sensitivity and function	L & Ds	PPT, IB	15 min
(b)	• Time, mercury, pressure, and thermal switches • Proximity switch operation principles • Rheostat function and relay basics • Installation safety precautions for switches and relays	L & Ds	PPT, PDF	20 min
(c)	• Review of switch types and applications • Safety protocols for installation procedures • Summary of key learning points	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a push switch and a toggle switch.
(b) Describe the safety precautions required when installing a relay.
(c) How does a thermal switch function in an avionics system?

12. Home Assignment and Follow Up	Prepare a brief note on the differences between a rheostat and a standard switch, referencing the next lesson on fuses and circuit breakers.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Fuses, current limiters, limiting resistors, circuit breakers, reverse current cut-out relays, reverse current circuit breakers Safeties to be taken installing fuses & circuit breakers. Practical on fuse & circuit breaker.	Refer to the master syllabus reference list	LP 13

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min